



University of Central Punjab

Faculty of Information Technology

PF Project

Matrix Manipulator

Project:

You are required to create a program that provides near-complete functionality over the matrix space. Users should be given a simple and smooth work flow that implements following details according to specs:

Front end:

- ✓ User can enter matrix personally or through files of their choosing
- ✓ User can get output on console or through files of their choosing
- ✓ Program should not end until user wishes to end their task
- ✓ User can enter Matrix of any size
- ✓ In case of file data entry each file will contain single matrix and you must use auto-grow techniques to read data

Functional Requirements:

- ✓ Sum of Matrix (sum of all values)
- ✓ Product of Matrix (product of all values)
- ✓ Row-wise Average
- ✓ Column-wise Average
- ✓ Average of whole Matrix

- ✓ Row-wise sorting of Matrix
- ✓ Column-wise sorting of Matrix
- ✓ Scalar Matrix Addition
- ✓ Addition of two Matrices
- ✓ Scalar Matrix Subtraction
- ✓ Subtraction of two Matrices
- ✓ Scalar Matrix Multiplication
- ✓ Multiplication of two Matrices
- ✓ Scalar Matrix Division
- ✓ Matrix transpose
- ✓ Matrix determinant
- ✓ Matrix Inverse

Back end:

- ✓ Use separate function for every task listed above that is controlled by menu function which guides users through the various options
- ✓ Use separate functions for helping tasks such as file input/output
- ✓ Use structs to represent Matrices
- ✓ Handle any and all sanity checks such as “size < 1”
- ✓ In case there are any operations the program cannot perform, show a proper error message on screen to user to clearly state what is wrong with the input
- ✓ Code should be thoroughly commented with appropriate details
- ✓ There should be no memory leakage at all throughout the program
- ✓ All code must be 100% generic

Useful links:

- ✓ <https://www.shelovesmath.com/algebra/advanced-algebra/matrices-and-solving-systems-with-matrices/>
- ✓ [https://en.wikipedia.org/wiki/Matrix_\(mathematics\)](https://en.wikipedia.org/wiki/Matrix_(mathematics))
- ✓ <https://www.khanacademy.org/math/algebra-home/alg-matrices>
- ✓ <https://www.mathsisfun.com/algebra/matrix-introduction.html>
- ✓ <https://courses.lumenlearning.com/boundless-algebra/chapter/introduction-to-matrices/>
- ✓ <http://mathworld.wolfram.com/Matrix.html>